

Non-Anticipative Hierarchical Event Diagnosis and Physics-Guided Candidate Localization from Sparse PMU Measurements

Jorge Luis Mayorga Taborda, Yessica Africano Rodriguez, Masoud Barati, Payman Dehghanian, and David Felipe Celeita Rodríguez

Abstract—Sparse phasor measurement unit (PMU) coverage turns event localization into an inverse source-inference problem: the initiating BUS or LINE is often unmeasured, while telemetry loss can resemble a physical transient. This paper studies three questions under a strict non-anticipative evaluation protocol: whether typed BUS/LINE/PMU routing changes the accuracy–resource trade-off, whether candidate-conditioned network information improves localization, and which event families remain difficult under an eight-PMU placement. The study uses 690 five-second ANDES simulations of the IEEE 39-bus system. Complete trajectories are separated across training, validation, and test partitions, and each timestamp-level decision uses only measurements available at or before that timestamp through a trailing 3-s representation. Flat, typed, and candidate-ranking ExtraTrees models are compared with the same trajectories, random seeds, validation procedure, and total 360-tree budget. Flat obtains 0.977 detection F1 and 0.535 row-level physical Top-1. Typed routing obtains 0.965 and 0.496, respectively, while reducing serialized size from 98.3 to 28.1 MB and model-only scoring time from 0.051 to 0.025 s per input minute. Its offline scenario-level physical Top-1 is 0.614 versus 0.590 for flat, with a paired interval that includes zero. The impedance-conditioned candidate ranker gives no aggregate localization gain and degrades line-outage Top-1 in the tested configuration. Missing-only detection and load localization remain failure cases. The results show that typed semantics can reduce computational cost without improving row-level accuracy, while a single static impedance descriptor is insufficient as a universal source representation. The reported estimates apply to new trajectories at known assets.

Index Terms—ANDES, event detection, event localization, ExtraTrees, phasor measurement units, physics-guided machine learning, synchrophasors.

I. INTRODUCTION

PHASOR measurement units provide time-synchronized voltage and current phasors together with frequency-related measurements, enabling wide-area observation of transmission-system dynamics [1], [2]. Localization is harder than event detection when only a small fraction of buses is instrumented:

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The initial prototype underlying this study was developed for the Synchrophasor Anomaly Detection Competition at the 5th International Conference on Smart Grid Synchronized Measurements and Analytics (SGSMA 2026), Santiago, Chile, June 1–4, 2026.

a disturbance at an unobserved bus or line must be inferred from the response measured elsewhere in the network. In the SGSMA 2026 problem, eight PMUs observe the IEEE 39-bus system while the required output includes event type and origin [3].

Sparse sensing creates two distinct ambiguities. Different physical sources can produce similar responses at the available PMUs, so state observability does not imply source identifiability [4], [5]. Separately, missing or corrupted telemetry can resemble the onset of a physical event unless the measurement process is modeled explicitly [6]–[8]. BUS, LINE, and PMU origins therefore should not be treated as interchangeable labels.

The competition prototype used complete 30-s windows, current and voltage phasors, ROCOF, robust statistics, multi-scale rolling features, adaptive residuals, and graph-temporal summaries. That design was appropriate for retrospective competition scoring, but a complete event window is not an appropriate information set for timestamp-level online inference. The formulation evaluated here is intentionally different: it uses a trailing 3-s history, 136 past-only variables, whole-trajectory train/validation/test separation, and one prediction per timestamp. Section III makes this transition explicit so that results from the competition prototype are not used to validate the non-anticipative formulation.

The experiments address three research questions. **RQ1** asks whether typed BUS/LINE/PMU routing improves the accuracy–resource trade-off relative to flat output heads. **RQ2** asks whether candidate-conditioned network information improves source localization under the current target-covered split. **RQ3** asks which event families remain difficult when only the eight prescribed PMUs are observed. The paper contributes a non-anticipative structured-output formulation, a matched comparison of flat, typed, and candidate-conditioned estimators, and an event-family analysis that separates resource savings, localization errors, and measurement-integrity failures. The physics-guided candidate model is evaluated as a complete design; the present experiment does not isolate the impedance descriptor by itself.

II. RELATED WORK AND POSITIONING

State estimation, missing-measurement reconstruction, and event localization use similar synchrophasor streams but estimate different latent quantities. Incomplete observability, communication latency, and synchronization errors therefore

affect them differently [9]. We focus on transmission-system event diagnosis and use state-estimation work where it clarifies the information that sparse synchronized measurements can retain.

A. Event Identification from PMU Streams

Integrated PMU diagnosis is well established. Pandey *et al.* combine detection, classification, localization, topology information, and bad-data processing [10]. Aalam and Shubhanga use wavelet energy and statistical indicators on multiple PMUs, including ISO-NE field records [11]. Long field records introduce drift and operating-condition variation: Yuan *et al.* use two years of PMU measurements and event logs [12], Ahmed *et al.* study drifting synchrophasor streams [13], and Cheng *et al.* develop an online GAN detector [14]. Limited labels have also been addressed through structural tensor learning [15].

Yuan, Wang, and Wang infer PMU-to-PMU interactions from data rather than fixing the transmission graph [16]. A learned interaction graph may be predictive without representing admittance, transfer impedance, or contingency sensitivity. The review in [17] spans network-aware features, model-based constraints, and equation-coupled learning; DynamoPMU, for example, extracts nonlinear dynamical structure from μ PMU trajectories [18]. These constructions make different assumptions about the grid and should not be grouped as equivalent uses of physics.

B. Sparse Localization, Placement, and Candidate Models

Sensor placement is part of the localization problem. Zhao, Goldsmith, and Poor select PMU locations so that projected line-outage signatures remain separated [19]. Li, Deka, Chertkov, and Wang combine CNN-based faulted-line localization with PMU placement on the IEEE 39- and 68-bus systems [20]. Jamei *et al.* analyze the statistical distinguishability of competing fault hypotheses under sparse sensing [5]. Yildiz and Abur study fault location with few PMUs and later optimize sparse PMU placement for event identification [21], [22].

Azizi *et al.* compare candidate generation-loss locations against network-predicted PMU responses [23] and use candidate-dependent network relationships again for line-outage monitoring [24]. Sánchez Cortés *et al.* use a related physical-consistency argument for generation-loss estimation under high renewable penetration [25]. The matched quantity is event dependent: an opened line changes network redistribution, whereas a generation loss appears through an injection imbalance and subsequent electromechanical response.

Matching Networks, Prototypical Networks, and Relation Networks likewise compare an observation with support examples or class representations [26]–[28]. Here the candidate is a physical BUS, LINE, or PMU asset, so the representation must carry enough network information to separate competing sources.

C. State and Measurement Reconstruction under Sparse Observation

Sparse PMUs are central to state estimation, but the latent target is different. Classical DSE combines dynamic models,

recursive estimation, and observability conditions [30]; Qi, Sun, and Kang connect PMU placement to DSE observability through an empirical Gramian [31]. Recent work augments this pipeline with graph and factor-graph message passing [32], [33], explicit network relations [34], and topology-aware GNNs for PMU-unobservable or degraded systems [35]. Missing PMU values can be reconstructed through tensor decomposition [36] or deep learning with magnitude-trend decoupling [37]. These methods recover electrical states or measurements rather than an event origin.

D. Generalization, Integrity, and Evaluation Scope

A trajectory holdout, an asset holdout, a sensing-layout change, and cross-network transfer answer different questions. Li, Ma, and Weng address cross-grid transfer by aligning event zones across different measurement spaces [29]; Wu *et al.* study zero-shot transfer across topology reconfigurations [38]. A new trajectory at a known asset, a source withheld from fitting, a new PMU layout, and a different network are four different evaluation units. The experiment reported here addresses only the first.

Telemetry failures change the observation process rather than the physical grid state. Liu *et al.* show that imperfect real-world PMU measurements materially affect event classification [7]; event-signature and PNNL data resources document both physical disturbances and data-quality problems [39]–[41]. A failed PMU changes the observation operator and may itself be the location to be inferred.

III. POWER-SYSTEM MODEL, DATA, AND ANDES DYNAMIC SIMULATION

A. From Competition Prototype to Non-Anticipative Evaluation

The journal experiment differs from the competition pipeline in both its information set and its evaluation unit. The competition method summarized complete 30-s windows with a large physics-aware feature bank; the formulation evaluated here issues one decision per timestamp from a trailing 3-s history. Table II records the exact change so that the retrospective competition results are not confused with the non-anticipative experiment.

The prototype scores document the lineage of the work. Its presentation also showed a qualitative incremental localization ablation for rolling, adaptive-residual, and graph-temporal feature blocks; exact bar values were not tabulated and are not reused as numerical evidence.

B. IEEE 39-Bus Network and PMU Placement

The study uses the 39-bus New England system, represented by the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $|\mathcal{V}| = 39$, a 100-MVA system base, 60-Hz nominal frequency, and 10 generator buses [42]. The SGSMA measurement layout places PMUs at

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \quad (1)$$

The eight streams are time aligned and provide three-phase voltage and current phasors, frequency, ROCOF, availability, and timestamp information [3]. The exact PMU-to-bus mapping

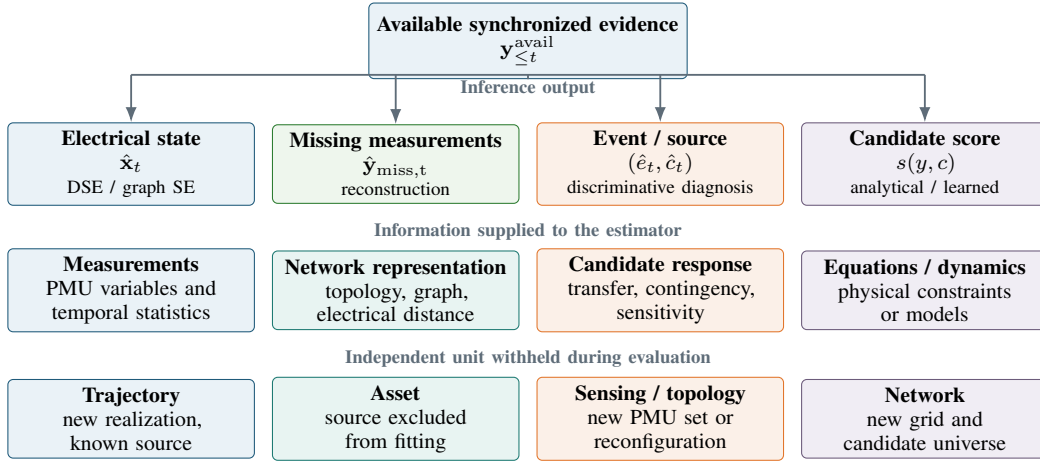


Fig. 1. Literature synthesis for sparse synchronized measurements. The first row separates four inference outputs; the lower rows distinguish how domain information enters the estimator and what experimental unit is held out.

TABLE I

REPRESENTATIVE EVENT-DIAGNOSIS LITERATURE AND THE PHYSICAL MEANING OF THE REPORTED OUTPUT. ACCURACY VALUES ARE NOT COMPARED BECAUSE EVENT INVENTORIES, CANDIDATE SPACES, AND SENSING LAYOUTS DIFFER.

Work	Output	System / sensing	Network or physical information	Evaluation unit
Zhao–Goldsmith–Poor [19]	line-outage signature	IEEE 14/24/30; PMU placement	projected outage-signature separation	placement / separability
Li <i>et al.</i> [20]	faulted line	IEEE 39/68; optimized PMUs	learned voltage signatures + placement	operating / sensing variations
Jamei <i>et al.</i> [5]	fault hypothesis	sparse-PMU fault study	information-theoretic distinguishability	placement / observability
Azizi <i>et al.</i> [23], [24]	generator or line candidate	transmission systems; sparse PMUs	candidate-specific network response	candidates in studied systems
Yuan <i>et al.</i> [12], [16]	event type	large real PMU corpora	temporal model; learned PMU interactions	temporal field data
Li–Ma–Weng [29]	event type + event zone	source and target grids	latent-domain / label-space alignment	cross-grid transfer
Yildiz–Abur [21], [22]	fault / event location	transmission test systems; few PMUs	CNN signatures + PMU placement	few-PMU / placement studies
This study	BUS / LINE / PMU origin + event type	IEEE 39; 8/39 PMUs	typed routing + candidate network descriptor	trajectory holdout; known assets

TABLE II

COMPETITION PROTOTYPE AND NON-ANTICIPATIVE FORMULATION EVALUATED IN THIS PAPER. HISTORICAL PROTOTYPE SCORES DOCUMENT THE PROJECT LINEAGE AND ARE NOT COMPARED STATISTICALLY WITH SEC. VII.

Element	Competition prototype	Evaluated formulation
Decision / context	complete 30-s window with pre/early/mid/late/full segments	one decision per timestamp from a trailing 3-s history
Signals	voltage/current magnitudes and angles, frequency, ROCOF, availability	voltage magnitude, unwrapped voltage angle, frequency, availability
Feature bank	robust deviation, derivatives, multiscale rolling, RLS residuals, phase/power proxies, graph-temporal summaries	current/reference departure, trailing mean/std./slope, missing fraction, fixed scalar summary
Dimension / network	45,162 archived variables; electrical-distance-weighted PMU pair features	136 variables; candidate-to-PMU impedance descriptor from Z_{bus}
Data / split	synthetic competition windows with RAW0001 transfer analysis	690 ANDES trajectories; 414/138/138 grouped train/validation/test
Historical scores	SIM: det. acc. .9693, cls. acc. .9673, M-F1 .9598, loc. Top-1 .8524; RAW0001: 1.000/1.000/1.000/.6667	reported results are the target-covered trajectory experiment of Sec. VII

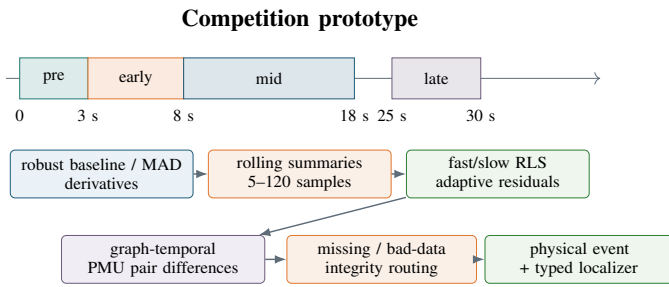


Fig. 2. Competition feature and routing logic reconstructed from the SGsMA presentation. The journal experiment preserves the event-dependent interpretation but replaces the complete-window bank with a smaller past-only representation.

is PMU1–Bus39, PMU2–Bus29, PMU3–Bus10, PMU4–Bus22, PMU5–Bus19, PMU6–Bus2, PMU7–Bus5, and PMU8–Bus6.

Figure 4 shows the electrical connectivity used by the simulation and the network descriptor. The drawing is schematic rather than geographic; solid edges denote transmission branches and dashed edges the generator step-up transformer connections represented in the standard case data.

Let $\mathbf{m}(t) = \mathbf{h}(\mathbf{x}(t), \mathbf{y}(t))$ denote the full network measurement vector implied by the dynamic and algebraic states. Sparse PMU sensing is written as

$$\mathbf{y}_t^{\text{PMU}} = H_{\mathcal{P}_t} \mathbf{m}(t) + \boldsymbol{\epsilon}_t, \quad (2)$$

where $H_{\mathcal{P}_t}$ selects the channels available at the PMU set $\mathcal{P}_t \subseteq \mathcal{P}$ and $\boldsymbol{\epsilon}_t$ represents the sampled measurement perturbation. Under normal telemetry $\mathcal{P}_t = \mathcal{P}$; a missing-data event removes one PMU block from the observation operator. The localization problem therefore acts on a projection of the network response, not on the full state.

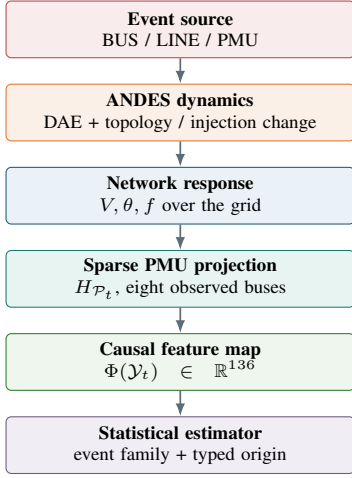


Fig. 3. End-to-end inference path from a physical or integrity event to the sparse PMU observation and statistical estimator.

C. Network Equations and Physical Event Signatures

At the network level, Kirchhoff's current law gives

$$\mathbf{i} = Y_{\text{bus}} \mathbf{v}, \quad (3)$$

and the complex power injection at bus i is

$$S_i = P_i + jQ_i = V_i \left(\sum_{j=1}^N Y_{ij} V_j \right)^*. \quad (4)$$

Writing $Y_{ij} = G_{ij} + jB_{ij}$ and $\theta_{ij} = \theta_i - \theta_j$ gives the familiar nonlinear AC balance

$$P_i = |V_i| \sum_j |V_j| (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}), \quad (5)$$

$$Q_i = |V_i| \sum_j |V_j| (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}). \quad (6)$$

These equations explain why a disturbance at an unobserved bus can still be inferred from correlated changes at the PMU buses.

For small current-injection perturbations around an operating point,

$$\Delta \mathbf{v} \approx Z_{\text{bus}} \Delta \mathbf{i}, \quad Z_{\text{bus}} = Y_{\text{bus}}^+, \quad (7)$$

which motivates the impedance-derived candidate descriptor used later. Electromechanical frequency changes follow active-power imbalance. A classical first-swing interpretation is

$$\frac{2H_i}{\omega_s} \dot{\omega}_i = \Delta P_{m,i} - \Delta P_{e,i} - D_i \Delta \omega_i, \quad \dot{\delta}_i = \Delta \omega_i, \quad (8)$$

so frequency and ROCOF carry information about generation/load imbalance [43]. Line outages are different: they redistribute branch flows and phase-angle differences. For interpretation, the post-outage flow change on monitored line ℓ due to outage m can be written in the standard sensitivity form

$$P_\ell^{\text{post}} \approx P_\ell^{\text{pre}} + \text{LODF}_{\ell,m} P_m^{\text{pre}}. \quad (9)$$

The implemented localizer does not use LODF features. Equation (9) shows why a single static impedance descriptor need not be equally informative for faults, line outages, and injection changes.

Figure 5 complements Table III with qualitative signal sketches. The integrity cases now use a more literal visual language inspired by the SGSMA presentation: missing data is shown as packet loss between sender and receiver, while bad data is shown as an otherwise smooth trajectory corrupted by an isolated outlier. These panels are not measured trajectories; they summarize the signal shapes emphasized in the slides and reused in the journal narrative to explain why different event families activate different feature groups.

D. ANDES Time-Domain Simulation and Data Construction

The dynamic trajectories are generated with ANDES on the IEEE 39-bus system [44]. ANDES represents the model as a differential-algebraic system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{u}; \boldsymbol{\eta}), \quad (10)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{u}; \boldsymbol{\eta}), \quad (11)$$

where $\mathbf{x}$ collects dynamic device states, $\mathbf{y}$ algebraic network variables, $\mathbf{u}$ disturbances, and $\boldsymbol{\eta}$ model parameters. Physical scenarios modify network or injection variables and propagate through the DAE solution. Missing and bad-data scenarios are applied to the sampled measurement stream so that telemetry failures remain distinct from physical grid disturbances.

All numerical results in Sec. VII use a separate 690-trajectory simulation set rather than the original 90-min competition stream. The simulation set contains 138 target keys with five replicas each. Every event begins at 2 s; faults last 0.2 s, persistent generation/load changes continue through the record, and missing/corrupted intervals last 0.7 s. Event severity, operating scale, and simulator seed vary across replicas. Scale, noise, and sampling rate are calibrated from normal rows in RAW0001; RAW samples are not used for model fitting or test-threshold selection.

The competition specification defines Event 6 generically as missing data concurrent with a physical event. In the simulation set used here, Event 6 is instantiated as generation change plus missing data, so its physical source is a BUS. The competition specification describes Event 8 as unknown/ambiguous; the simulation set instantiates it as a closed mixed physical class and does not evaluate open-set recognition. These distinctions keep the reported estimand tied to the actual simulated labels.

The simulated outputs are sampled trajectories, not a complete standard-compliant PMU/PDC chain. Time-synchronization error, packet reordering, communication buffering, estimator response, and other instrument-level effects are not reproduced.

IV. PROBLEM FORMULATION AND EVALUATION UNIT

A. Structured Output and Candidate Spaces

At timestamp t , the required output is

$$\hat{\mathbf{z}}_t = \left(\hat{e}_t, \hat{\ell}_t^{\text{phys}}, \hat{\ell}_t^{\text{int}} \right), \quad (12)$$

where either origin is empty when it does not apply. The simulation set contains 39 fault BUS candidates, 34 LINE candidates, 10 generation BUS candidates, 18 load BUS candidates, and eight PMU integrity sites. The 34 line-outage

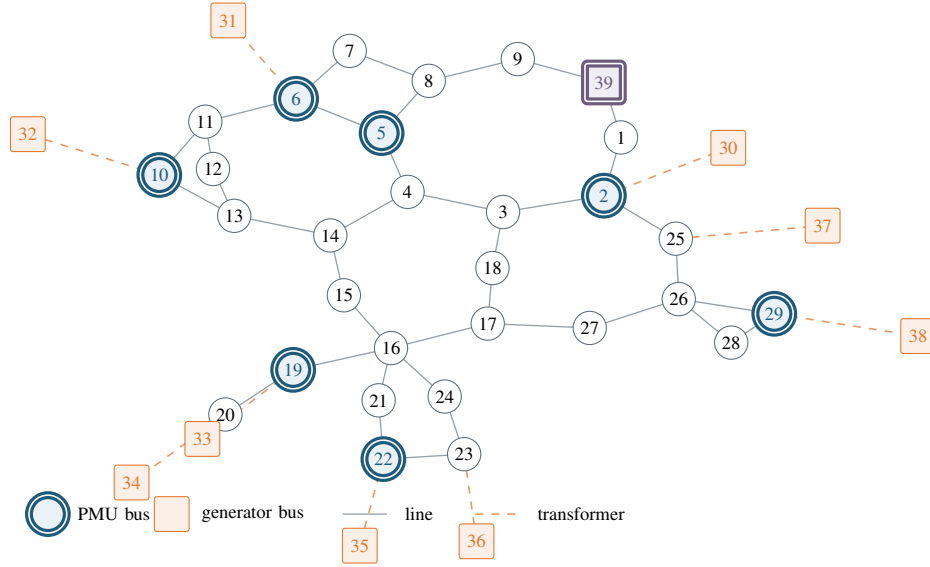


Fig. 4. Schematic one-line connectivity of the IEEE 39-bus system used in the study. Double-outline nodes mark the eight PMU buses; rectangular nodes are generator buses. Connectivity follows the standard case39 branch data; the layout is not geographic.

TABLE III
CHALLENGE EVENT FAMILIES, THEIR PHYSICAL INTERPRETATION, AND HOW THE COMPETITION PROTOTYPE AND THE NON-ANTICIPATIVE EXPERIMENT REPRESENT THEM.

Event	Physical mechanism	Prototype cues	Causal encoding used here	Origin / network relation
1 Fault	short-circuit current and voltage depression	voltage span/derivative, robust z , early-segment change, cross-PMU severity	$ V $ current, departure, mean, std., slope; θ response	BUS; transfer-impedance / fault-response relation
2 Line outage	branch-flow redistribution after topology change	current span/derivative, current-angle change, graph-temporal pair differences	voltage-angle and frequency statistics; no current features in the present 136-D vector	LINE; PTDF/LODF or post-contingency response
3 Generation	active-power imbalance and electromechanical response	frequency shift, ROCOF, P/Q proxy, sustained dynamics	frequency and angle departure/mean/slope; no ROCOF in the present vector	BUS; injection sensitivity + machine dynamics
4 Load	active/reactive injection change	sustained current shift, active-power proxy, rolling energy	$ V $, θ , and f departure/mean/slope	BUS; AC/Jacobian injection sensitivity
5 Missing	observation loss without grid-state change	data-present fraction, NaN fraction, time-stamp gaps, missing-run duration	trailing missing fraction $m_p(t)$ formed before imputation	PMU; time-varying observation operator
6 Missing + phys.	simultaneous telemetry loss and physical disturbance	missing gate plus physical-event evidence, then typed routing	availability pattern plus the same physical variables used by the routed BUS/LINE head	PMU for integrity + BUS/LINE for the concurrent physical source
7 Bad data	isolated inconsistent telemetry	single-channel robust anomaly, phase inconsistency, top/second anomaly ratio	availability plus the fixed scalar disturbance summary and cross-signal statistics	PMU; measurement-integrity model
8 Unknown / mixed	physical evidence without a clean match to Events 1–4	power proxy, rolling dynamics, adaptive residuals, graph-temporal pattern	off-manifold combination of the same causal statistics, captured through low-confidence alternatives	BUS; unresolved physical source

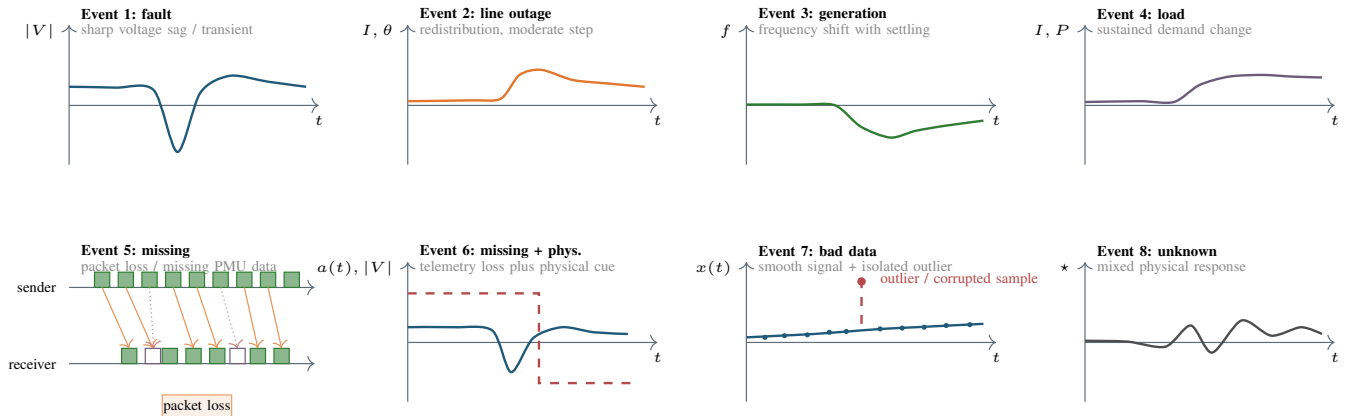


Fig. 5. Qualitative PMU-level signatures for the abnormal challenge classes. The panels condense the shapes highlighted in the SGSMA presentation: faults produce sharp voltage sags, line outages redistribute currents/angles, generation and load events leave sustained frequency or power-demand signatures, missing and bad-data events primarily alter the observation process, and Event 8 collects mixed physical responses.

TABLE IV
SIMULATION-SET INVENTORY. EACH TARGET KEY HAS FIVE REPLICAS.

e	Family	Keys	Scenarios
0	normal	1	5
1	bus fault	39	195
2	line outage	34	170
3	generation change	10	50
4	load change	18	90
5	missing data	8	40
6	generation + missing	10	50
7	bad data	8	40
8	mixed physical class	10	50
Total		138	690

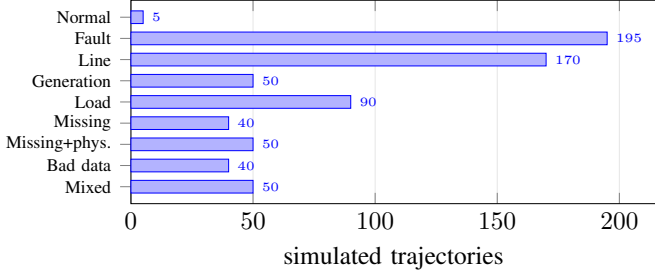


Fig. 6. Scenario composition of the 690-trajectory ANDES set. Scenario counts should not be interpreted as row-level class balance because event durations differ across families.

TABLE V
MEASUREMENT CONTRACT AND CAUSAL INPUT SET.

Measurement family	Available	Used
Voltage magnitude / angle	yes	yes
Current magnitude / angle	yes	no
Frequency	yes	yes
ROCOF	yes	no
Availability / missingness	yes	yes
Timestamp	yes	alignment/windowing

TABLE VI
EVENT-COMPATIBLE ORIGIN SPACES USED IN THE EXPERIMENT.

e	Event	Physical candidates	Integrity
0	normal	—	—
1	fault	39 BUS	—
2	line outage	34 LINE	—
3	generation change	10 BUS	—
4	load change	18 BUS	—
5	missing data	—	8 PMU
6	generation + missing	10 BUS	8 PMU
7	bad data	—	8 PMU
8	mixed physical class	10 BUS	optional

candidates are the physical transmission branches in the case39 branch table with unit tap and zero phase shift; transformer branches are excluded from the line-outage label space.

Event 6 is instantiated in this simulation set as generation change plus missing data, so its physical origin is a BUS and its integrity origin is a PMU. Event 8 is evaluated as a closed mixed physical class rather than an open-set class.

B. Non-Anticipative Observation Contract

Let

$$\mathcal{Y}_t = \{\mathbf{y}_{\mathcal{P}_\tau, \tau} : \tau \leq t\} \quad (13)$$

denote the measurements available up to decision time t , with $\mathcal{P}_\tau \subseteq \mathcal{P}$ under dropout. The estimator is non-anticipative when

$$\hat{\mathbf{z}}_t = f(\Phi(\mathcal{Y}_t)) \quad (14)$$

is unchanged by any modification to samples after t .

If ρ_t is the abnormal forest score, the first-stage decision is

$$\hat{d}_t = \mathbf{1}\{\rho_t \geq \gamma\}, \quad (15)$$

where γ is selected on validation data and frozen before test evaluation. Once an event family is selected, localization is restricted to its compatible candidate set. A fixed class head predicts within that set, while the candidate-conditioned variant returns

$$\hat{c}_t = \arg \max_{c \in \mathcal{C}_{\hat{e}_t}} s_\theta(c | \mathbf{x}_t). \quad (16)$$

Every target asset appears in training, validation, and test. The independent test unit is therefore a new trajectory at a known source. This split does not estimate performance for an unseen BUS or LINE, a new PMU layout, or another network.

V. HIERARCHICAL STATISTICAL METHOD AND NETWORK-CONDITIONED CANDIDATES

A. Causal Feature Construction and Missing-Data Preprocessing

At timestamp t , the extractor reads the trailing interval

$$\mathcal{W}_t = [\max(t - 3 \text{ s}, t_0), t]. \quad (17)$$

For retained signal $s \in \{|V|, \theta, f\}$ at PMU p , the first finite value defines a causal scenario reference

$$b_{p,s} = x_{p,s} \left(t_{p,s}^{(0)} \right). \quad (18)$$

Each signal contributes

$$\phi_{p,s}(t) = [x_{p,s}(t), x_{p,s}(t) - b_{p,s}, \mu_{p,s}(t), \sigma_{p,s}(t), \beta_{p,s}(t)], \quad (19)$$

where μ , σ , and β are the trailing mean, standard deviation, and least-squares slope over $\mathcal{W}_t$. A trailing missing fraction $m_p(t)$ and a deterministic scalar disturbance summary $q_p(t)$ complete the PMU block,

$$\mathbf{x}_{p,t} = [\phi_{p,|V|}(t), \phi_{p,\theta}(t), \phi_{p,f}(t), m_p(t), q_p(t)] \in \mathbb{R}^{17}, \quad (20)$$

and

$$\mathbf{x}_t = [\mathbf{x}_{p,t}]_{p \in \mathcal{P}} \in \mathbb{R}^{136}. \quad (21)$$

The scalar summary is fixed across all compared models.

Missingness is preserved before numerical filling. The availability/NaN indicators are formed first; remaining non-finite feature values presented to the tree models are median-imputed using statistics estimated on the training partition. This follows the tabular preprocessing logic of the original tree pipeline while preserving missingness as predictive evidence. A future-change unit test modifies samples strictly after t and verifies that all features at $\tau \leq t$ remain unchanged.

The five statistics play complementary roles. Reference departure and trailing mean represent persistent level changes, trailing spread responds to transient variability, and the local slope represents ramps or settling dynamics. The missing fraction is computed before imputation so that the model retains

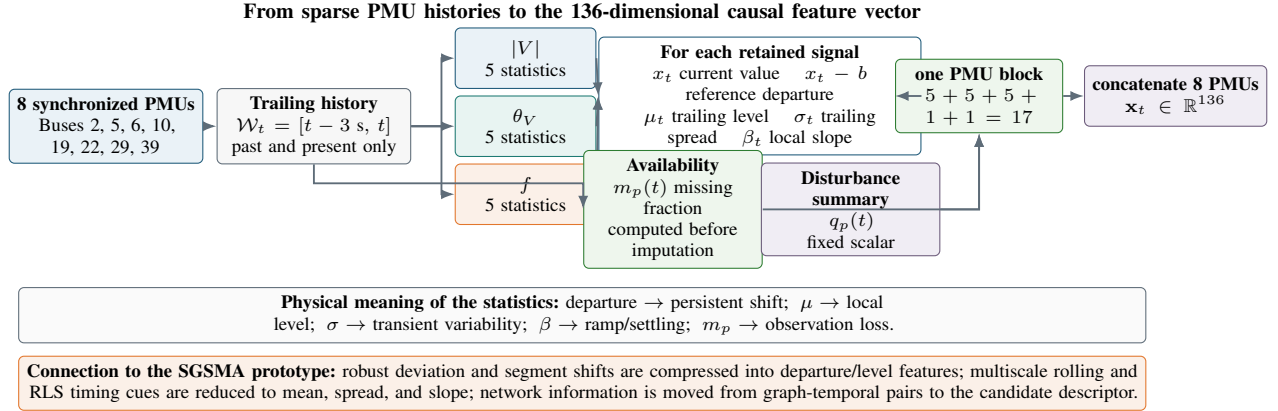


Fig. 7. Causal feature construction used by the journal experiment. The figure makes explicit how the richer feature-engineering logic of the competition prototype is reduced to a timestamp-feasible representation without future samples.

explicit evidence about the observation process. All imputation statistics are estimated from the training partition only. In the vocabulary of the competition prototype, departure/mean carry the sustained level-change information, standard deviation carries transient spread, slope carries part of the rolling/RLS timing information, and missing fraction carries the data-quality route. This mapping is shown explicitly in Fig. 7.

The competition prototype used robust deviation, sustained-versus-isolated anomaly shape, multiscale rolling summaries, adaptive residuals, and graph-temporal propagation cues. The non-anticipative formulation retains the underlying ideas of level change, transient spread, slope, missingness, and network-relative evidence, but reduces the representation so that each reported output can be computed without future samples.

B. ExtraTrees Statistical Learner

The learning layer uses Extremely Randomized Trees [45]. At each node, ExtraTrees evaluates random feature/threshold proposals and selects a split according to impurity reduction. If π_k denotes the class fraction in a node, the Gini impurity is

$$G = 1 - \sum_k \pi_k^2. \quad (22)$$

For tree m , let $p_m(k | \mathbf{x})$ be the class fraction in the reached leaf. The ensemble vote is

$$p_{\text{ET}}(k | \mathbf{x}) = \frac{1}{M} \sum_{m=1}^M p_m(k | \mathbf{x}). \quad (23)$$

The vote fractions are used for ranking and thresholding only.

Figure 9 makes the decision order explicit. The flat architecture fits unrestricted event, physical-location, and integrity-location heads. The typed architecture separates abnormal detection from event identification, then routes the predicted event to a BUS, LINE, or PMU origin space before localization. The candidate-ranking architecture preserves the same detector/event stage but replaces the fixed physical-location vocabulary with a score over explicit candidate assets. All three variants use the same trajectory partitions and total 360-tree budget.

The typed design preserves the prototype's separation between measurement-integrity failures and physical origins.

Missing or corrupted telemetry is assigned to a PMU origin space, whereas physical events are routed to BUS or LINE candidates. ExtraTrees is used to learn interactions among the engineered statistics without requiring a manually tuned rule for every event family.

C. Head-Specific Targets and Training Rows

The hierarchy uses different labels for different statistical questions. The abnormal detector is trained with

$$y_t^{\text{det}} = \mathbf{1}\{e_t \neq 0\}. \quad (24)$$

The event head uses the event family e_t , while a typed location head is fitted only on target-active rows whose event family maps to that origin type. Thus fault, generation, load, and the mixed physical class contribute BUS targets; line outages contribute LINE targets; and missing/bad-data rows contribute PMU integrity targets. Event 6 contributes both a physical BUS target and a PMU integrity target in the simulated bank.

For candidate-conditioned localization, each target-active row $\mathbf{x}_t$ is expanded into candidate rows. The true source receives label one and sampled non-target candidates receive zero. The candidate identity therefore enters through its type and network descriptor rather than through a fixed multiclass output index.

D. Network Descriptor and Candidate Interaction

From Eq. (3), the topology descriptor is built as

$$Z_{\text{bus}} = Y_{\text{bus}}^+, \quad d_{ab}^Z = |Z_{aa} + Z_{bb} - Z_{ab} - Z_{ba}|. \quad (25)$$

For a BUS candidate c ,

$$\psi_Z(c) = [d_{cp}^Z]_{p \in \mathcal{P}}. \quad (26)$$

For a LINE candidate $\ell = (i, j)$,

$$d^Z(\ell, p) = \min\{d_{ip}^Z, d_{jp}^Z\}, \quad \psi_Z(\ell) = [d^Z(\ell, p)]_{p \in \mathcal{P}}. \quad (27)$$

A fixed training-side normalization operator $\mathcal{S}_Z$ gives $\tilde{d}^Z = \mathcal{S}_Z(d^Z)$, and the PMU-wise interaction block is

$$\mathbf{x}_{c,t}^Z = \left[\mathbf{x}_{p,t} \exp(-\tilde{d}^Z(c, p)) \right]_{p \in \mathcal{P}}. \quad (28)$$

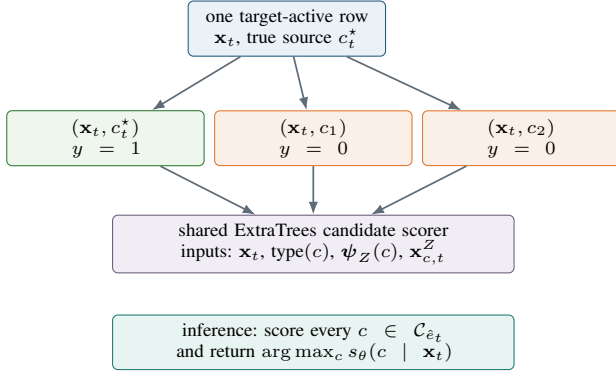


Fig. 8. Candidate-row construction. A physical training row is paired with its true source and a frozen set of admissible non-target candidates. All candidates are scored at inference.

For each target-active training row with true source c_t^* , the candidate label is

$$y_{c,t} = \mathbf{1}\{c = c_t^*\}. \quad (29)$$

Training uses the true candidate and a frozen subset of non-target candidates from the same admissible set; all admissible candidates are scored at inference. The scorer is

$$s_\theta(c | \mathbf{x}_t) = h_\theta(\mathbf{x}_t, \text{type}(c), \psi_Z(c), \mathbf{x}_{c,t}^Z), \quad (30)$$

with

$$\hat{c}_t = \arg \max_{c \in \mathcal{C}_{e_t}} s_\theta(c | \mathbf{x}_t). \quad (31)$$

The descriptor is physics guided rather than equation constrained. It embeds the network into the candidate representation but does not force the learner to satisfy the AC equations or the ANDES DAE during fitting. Because candidate-row construction and representation change together with the impedance variables, the current comparison evaluates the complete candidate-ranking design, not d^Z in isolation.

VI. EXPERIMENTAL DESIGN

A. Grouped Split and Statistical Controls

For every target key, three replicas enter training, one validation, and one test, yielding 414/138/138 whole-trajectory partitions. No trajectory contributes rows to more than one partition. The median imputer, tree ensembles, candidate-row construction, and training-side network-scaling constants are fitted or frozen from the training partition. Validation is used to select the abnormal decision threshold; the test trajectories are not accessed until final scoring. Because each target appears in every partition, the estimand is new-trajectory performance at known assets.

All variants use ExtraTrees with maximum depth 14, at least two samples per leaf, square-root feature subsampling, and a total budget of 360 trees. Equal total tree count controls one resource dimension but does not equalize per-head capacity because the architectures contain different numbers of heads.

ExtraTrees is not optimized by gradient descent and therefore has no epoch-wise loss trajectory. The appropriate learning diagnostics are score versus training-set size and score versus number of trees. Those sweeps are not inferred from the

fixed 360-tree experiment and are requested explicitly in the reproducibility/experiment specification delivered with this manuscript.

The abnormal decision threshold is selected once from a 33-value validation grid spanning 0.1–0.9 and is then frozen for the test set. The resulting thresholds are 0.75 for flat and 0.45 for typed in all three model seeds.

B. Primary Estimands

The primary online quantities are abnormal-event F1 and row-level physical Top-1. Event macro-F1 and physical Top-3 are secondary predictive metrics. Scenario Top-1 is an offline modal summary over target-active rows rather than a timestamp-level decision. Operational metrics are alarm episodes per normal-labeled minute and aligned-sample detection delay; resource metrics are serialized size, node count, and model-only scoring time.

C. Metrics and Uncertainty

Detection uses abnormal precision, recall, F1, row false-positive fraction, alarm episodes per normal-labeled minute, scenario detection, and aligned-sample detection delay. Event identification uses macro-F1 over labels 0–8. Localization uses row-level Top-1/Top-3 exact recovery. Scenario Top-1 is the modal physical prediction over target-active rows and is reported only as an offline aggregation; empty outputs count as errors.

Means and sample standard deviations summarize three model seeds. Paired intervals use 5000 bootstrap resamples of test scenarios within event family while preserving method-seed pairing. The intervals condition on this target-covered split; they do not represent uncertainty over held-out assets, PMU placements, or networks.

The candidate-support audit gives complete support for every evaluated typed candidate set because all target keys are present in training. Electrical-distance localization error is not available in the current result set, so the architecture comparison uses Top-1 and Top-3 physical-location accuracy.

The official SGSMA reporting guide also requests a full event confusion matrix and per-class metrics [3]. Those diagnostics require the locked row-level predictions from the causal experiment and are not reconstructed from aggregate scores. The accompanying experiment specification instructs the repository runner to export them together with threshold, ranking, importance, and learning-curve diagnostics.

D. Timing

Model-only timing was measured on Windows 11 with an Intel Core Ultra 9 185H, 16 physical and 22 logical cores, and 31.5 GB RAM using `n_jobs=-1`. Each seed was timed once without warm-up. The measurement excludes ingestion, alignment, feature computation, alarm aggregation, and output, and is interpreted only as descriptive batch-scoring time.

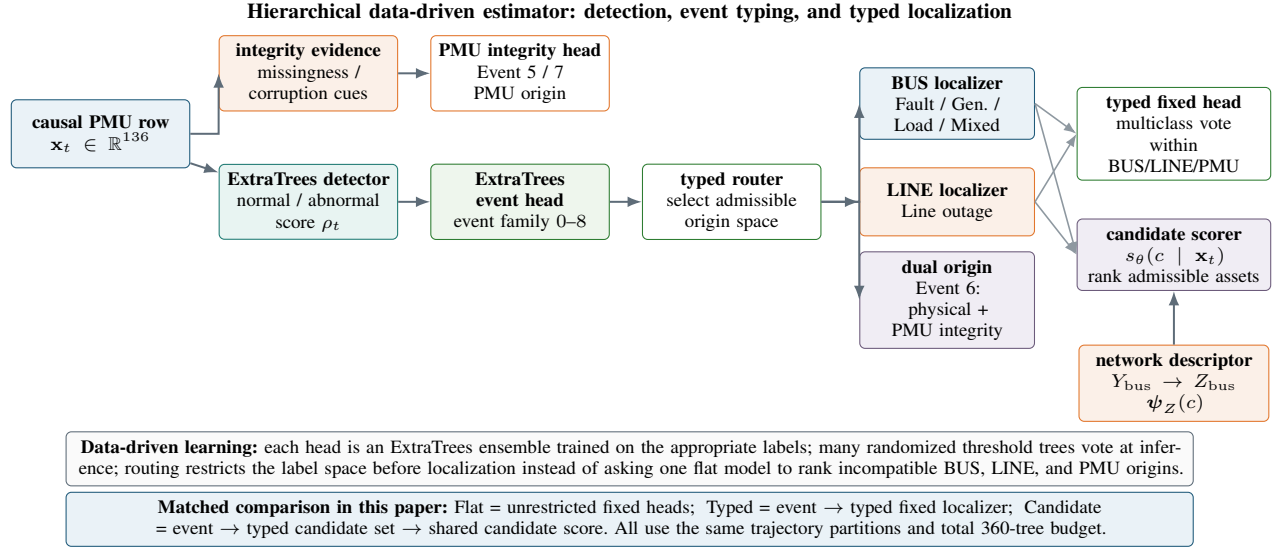


Fig. 9. Hierarchical tree-based inference architecture. The decision order follows the integrity-first and typed-localization logic of the SGSMA competition method, while the journal formulation uses the 136-dimensional past-only feature vector and adds an optional network-conditioned candidate scorer.

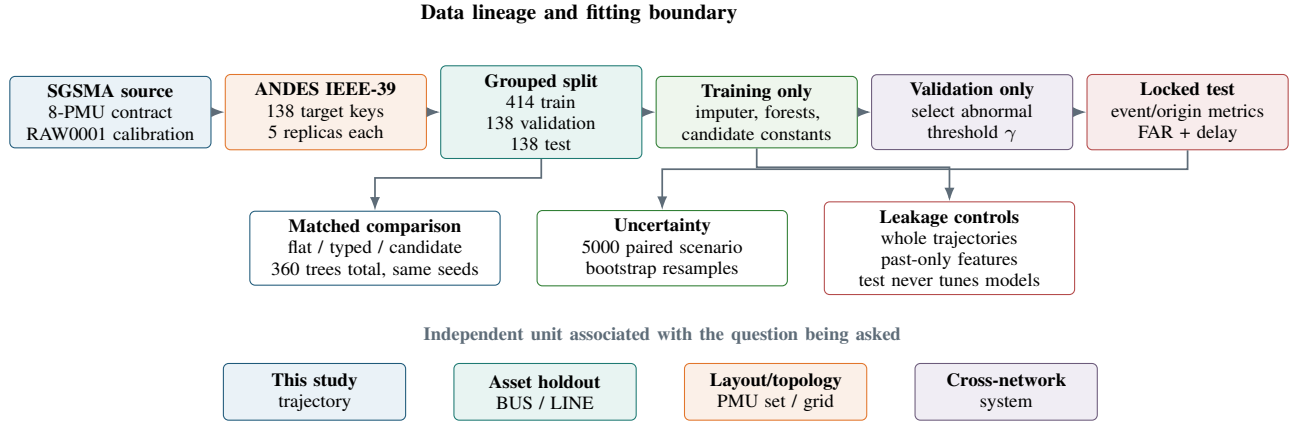


Fig. 10. Data lineage, model-fitting boundary, and evaluation units. RAW0001 informs the measurement-layer calibration, while the reported predictive metrics come from grouped ANDES trajectories. Model fitting is restricted to training, validation selects the abnormal threshold, and test trajectories remain locked until final scoring.

VII. RESULTS

A. Detection, Event Identification, and Localization

Flat obtains 0.995 ± 0.0004 abnormal precision, 0.959 ± 0.001 recall, and 0.977 ± 0.001 F1. Typed obtains 0.981 ± 0.001 , 0.951 ± 0.002 , and 0.965 ± 0.002 , respectively. The paired interval for typed minus flat detection F1 is $[-0.0207, -0.0045]$.

Row-level physical Top-1 is 0.535 ± 0.012 for flat and 0.496 ± 0.017 for typed. The paired interval is $[-0.0772, -0.0029]$. Offline scenario aggregation gives 0.614 ± 0.029 for typed and 0.590 ± 0.017 for flat; the interval $[-0.0551, 0.1019]$ includes zero. The candidate ranker obtains 0.462 physical Top-1 by row and 0.581 by scenario. Relative to typed, the paired intervals are $[-0.0734, 0.0047]$ and $[-0.1129, 0.0441]$.

B. Resource Trade-Off and Detection Delay

The serialized estimator bundle occupies 98.3 MB for the flat architecture and 28.1 MB for the typed architecture, a 71% reduction; model-only scoring time decreases from 0.051 to

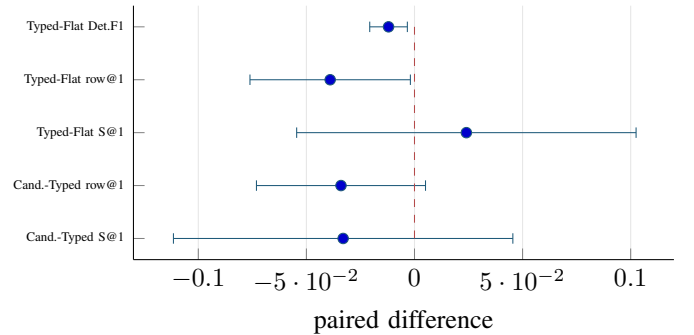


Fig. 11. Paired method differences already reported in the text. Horizontal intervals summarize the reported scenario-resampling ranges; intervals crossing zero indicate an inconclusive direction in this experiment.

0.025 s per input minute. The same 360-tree total is distributed across a different number of heads, so this is an accuracy–resource comparison rather than an equal-capacity comparison. Candidate expansion raises model-only scoring time to 0.586 s per input minute.

Flat detects 130 of 137 abnormal test scenarios with a

TABLE VII

TARGET-COVERED TEST RESULTS ACROSS THREE MODEL SEEDS. S@1 IS OFFLINE MODAL PHYSICAL TOP-1 PER SCENARIO; FAR IS ALARM EPISODES PER NORMAL-LABELED MINUTE. TIMING EXCLUDES FEATURE COMPUTATION.

Method	Det. P	Det. R	Det. F1	Event M-F1	Phys. S@1	FAR	Phys. @1/@3	Integ. @1/@3	s/min	Nodes (k)	Size (MB)
Flat	.995 ± .000	.959 ± .001	.977 ± .001	.800 ± .006	.590 ± .017	3.22 ± .14	.535/.671	.888/.888	.051	282	98.3
Typed	.981 ± .001	.951 ± .002	.965 ± .002	.791 ± .011	.614 ± .029	4.01 ± .40	.496/.646	.870/.874	.025	176	28.1
Candidate ranker	.981 ± .001	.951 ± .002	.965 ± .002	.791 ± .011	.581 ± .029	4.01 ± .40	.462/.622	.874/.876	.586	290	25.7

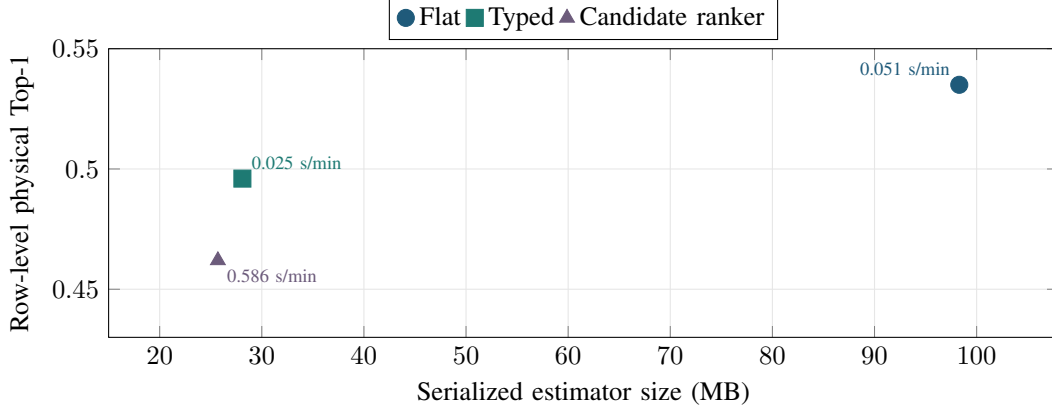


Fig. 12. Accuracy–resource trade-off for the three evaluated architectures. Typed routing reduces serialized size and model-only scoring time relative to flat heads, but the row-level localization point estimate is lower. Candidate expansion is compact on disk but slower at inference because each timestamp is expanded over candidate rows.

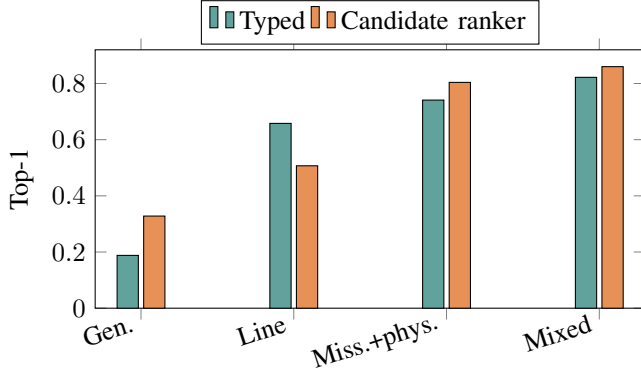


Fig. 13. Event-family physical Top-1 for typed class heads and the candidate-ranking design. The point estimates increase for generation, missing-plus-physical, and mixed events and decrease for line outages; the comparison does not isolate the impedance descriptor from the changed candidate representation.

mean aligned-sample delay of 0.0189 ± 0.0008 s; typed detects 129 with 0.0287 ± 0.0035 s. These values exclude communication, PMU/PDC, feature-computation, and alarm-aggregation latency.

C. Event-Family Behavior and Failure Modes

Typed labels every missing-only test row as normal, while flat recall for that class is 0.006 ± 0.006 . Load localization remains below 0.05 for all three methods. With 18 admissible load buses, uniform Top-1 selection gives approximately $1/18 = 0.056$, placing the load result near chance.

Relative to typed class heads, the candidate-ranking design changes generation Top-1 from 0.188 to 0.328, missing-plus-physical from 0.741 to 0.804, mixed-event physical Top-1 from 0.822 to 0.860, and line-outage Top-1 from 0.658 to 0.507. Line outages account for 3128 of the 7915 physical test rows.

VIII. DISCUSSION

A. RQ1: What Does Typed Routing Change?

Typed routing does not improve the principal row-level localization metric in this experiment. Flat achieves 0.535 physical Top-1 versus 0.496 for typed, and flat also has higher detection F1. The measurable benefit is structural and computational: incompatible BUS, LINE, and PMU origins are never compared inside one location head, serialized size falls from 98.3 to 28.1 MB, and model-only scoring time falls from 0.051 to 0.025 s per input minute. Typed routing is therefore an accuracy–resource and output-semantic trade-off, not an accuracy improvement.

B. RQ2: Does the Network-Conditioned Candidate Design Help?

Not in aggregate under the target-covered split. Candidate ranking gives 0.462 row-level physical Top-1 and 0.581 offline scenario Top-1, with paired intervals that do not show an advantage over typed class heads. The family-level results are more informative: generation, missing-plus-physical, and the mixed class have higher point estimates, whereas line-outage Top-1 falls from 0.658 to 0.507. This pattern is consistent with the physical mismatch already visible in Section III: a static bus-impedance proximity is closer to an injection-to-voltage relation than to a line-outage redistribution signature. Because candidate row construction changes together with the network descriptor, the experiment cannot attribute these differences to d^Z alone.

C. RQ3: Which Cases Remain Difficult?

Two failures dominate. First, the typed model labels all missing-only rows as normal even though missingness is

explicitly represented. This indicates that integrity routing should not rely solely on the learned abnormal score when deterministic availability information is present. Second, load localization is below 0.05 Top-1, close to the $1/18 = 0.056$ uniform baseline. The current results do not distinguish whether this is caused by weak features, insufficient operating-point variation, or poor source separability under the eight-PMU placement.

The prototype-to-journal comparison suggests two concrete hypotheses for these failures. The competition feature bank contained explicit current/angle redistribution cues for line outages, ROCOF and power-proxy cues for generation/load changes, and deterministic missing-data routing. Those cues are absent or compressed in the present 136-variable representation. The current experiment therefore cannot separate a limitation of sparse sensing from a limitation introduced by the reduced feature map; the planned current/ROCOF ablation and availability gate directly test that distinction.

The alarm rate is also too high for an operator-facing interpretation: flat and typed produce more than three alarm episodes per normal-labeled minute. Any deployment study would need persistence or hysteresis logic and must report the resulting FAR–recall–delay trade-off. The timing numbers reported here cover model scoring only and exclude feature computation, communication, PMU/PDC processing, and alarm aggregation.

IX. LIMITATIONS AND FUTURE WORK

The current experiment has four hard boundaries. Every source asset appears during fitting; all events begin at 2 s; one network and one PMU placement are used; and the simulator output does not reproduce a complete PMU/PDC communication chain. Three model seeds on one grouped split quantify model variation only, not target-, placement-, or network-level uncertainty.

Several follow-up tests are necessary before a stronger source-transfer or physics claim can be made. A fixed candidate scorer should compare absent, correct, and deliberately permuted network descriptors while holding candidate rows, negatives, tuning, and compute constant. Event-conditioned descriptors should then replace the universal static distance: transfer-impedance features for faults, PTDF/LODF-type signatures for line outages, and operating-point-dependent injection sensitivities for generation and load events. Leave-BUS-out and leave-LINE-out folds are required to test the principal advantage of candidate scoring, namely emission of a physical source excluded from fitting.

The measurement model also deserves a controlled expansion. Current phasors and ROCOF are present in the underlying PMU contract and were used by the competition prototype, but they are absent from the 136-variable formulation evaluated here. Their contribution should be measured non-anticipatively rather than assumed. Randomized event onset, broader operating points, multiple PMU layouts, longer normal streams, and a second transmission network are needed for external-validity claims. A deterministic availability gate should be tested for Events 5 and 6, and future localization reports should include

per-event confusion, electrical-distance error, and oracle-event routing in addition to Top-1/Top-3.

X. CONCLUSION

This study evaluates sparse-PMU event diagnosis under a non-anticipative timestamp-level information contract. The 690-trajectory ANDES experiment shows that typed BUS/LINE/PMU routing primarily changes model structure and resource use: it substantially reduces serialized size and model-only scoring time, but it does not improve row-level detection or localization relative to flat heads. The candidate-conditioned network representation likewise provides no aggregate gain under the target-covered split and exhibits a clear event dependence, with the largest degradation on line outages. Missing-only detection and load localization expose additional weaknesses that are hidden by aggregate scores.

The main empirical lesson is narrower than a universal “physics helps” claim. Sparse source inference depends on the event mechanism, the available observation set, and the unit held out during evaluation. Establishing a stronger physics-guided localization result requires an isolated descriptor ablation and asset-disjoint tests; establishing operational value additionally requires a realistic alarm budget and a fuller measurement chain.

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